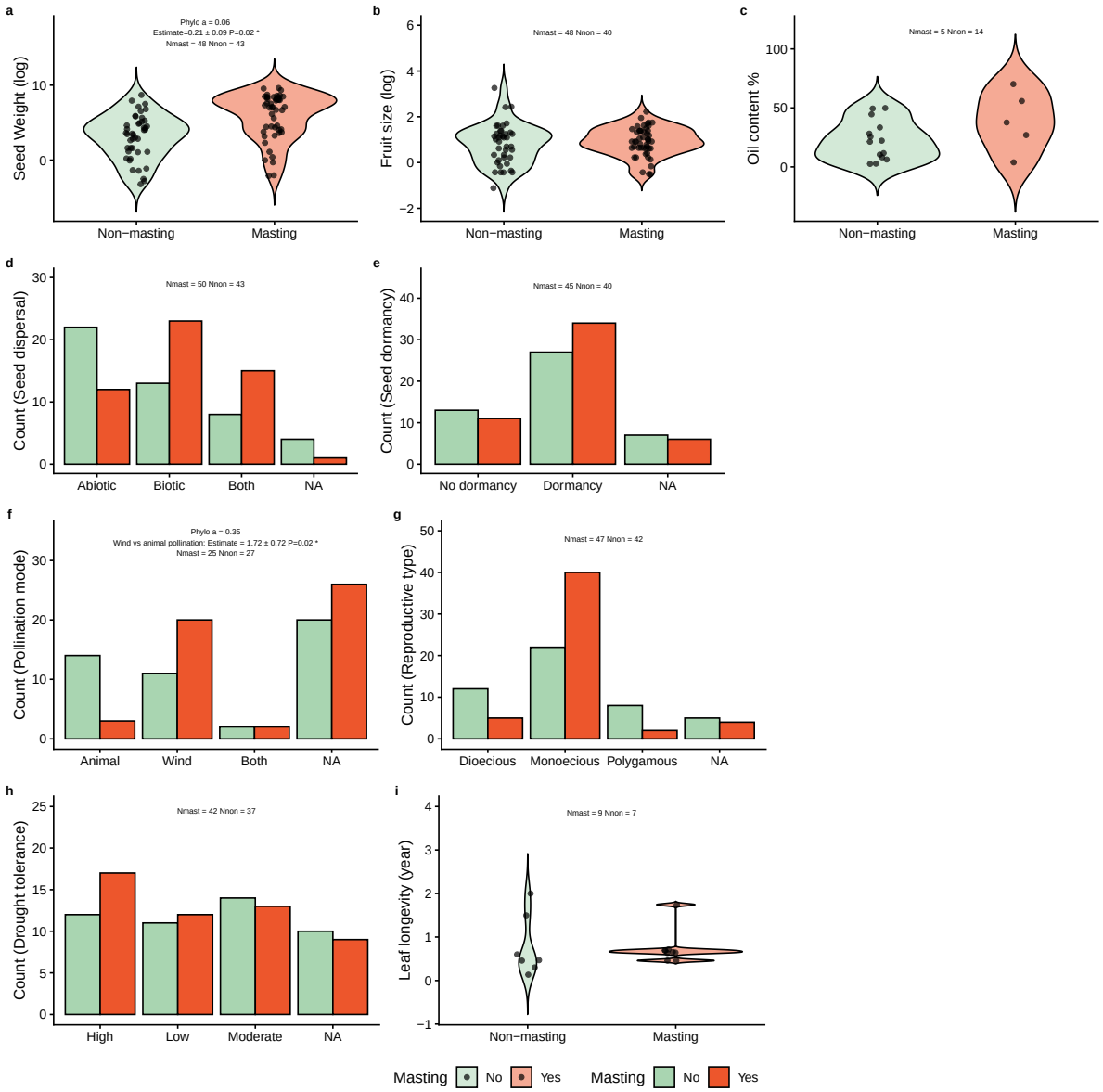

Figure 1: Relationships between plant traits (a) Seed weight; (b) Fruit size; (c) Oil content; (d) Dispersal mode; (e) Seed dormancy; (f) Pollination mode; (g) Reproductive mode; (h) Drought tolerance; (i) Leaf longevity and masting based on phylogenetic logistic regression for angiosperm species. Raw data are shown for each trait: continuous traits are displayed as violin plots, and categorical traits as histograms. Overlaid model summaries are derived from phylogenetic logistic regression models fitted using phyloglm. For each model, we report sample size (N), regression coefficient (Estimate), phylogenetic signal parameter (α), and associated P-value. (a)-(e) are traits related to predation satiation hypothesis; (f), (g) are traits related to pollination coupling hypothesis; (h), (i) are traits related to resource matching hypothesis. Legends are shared across panels.

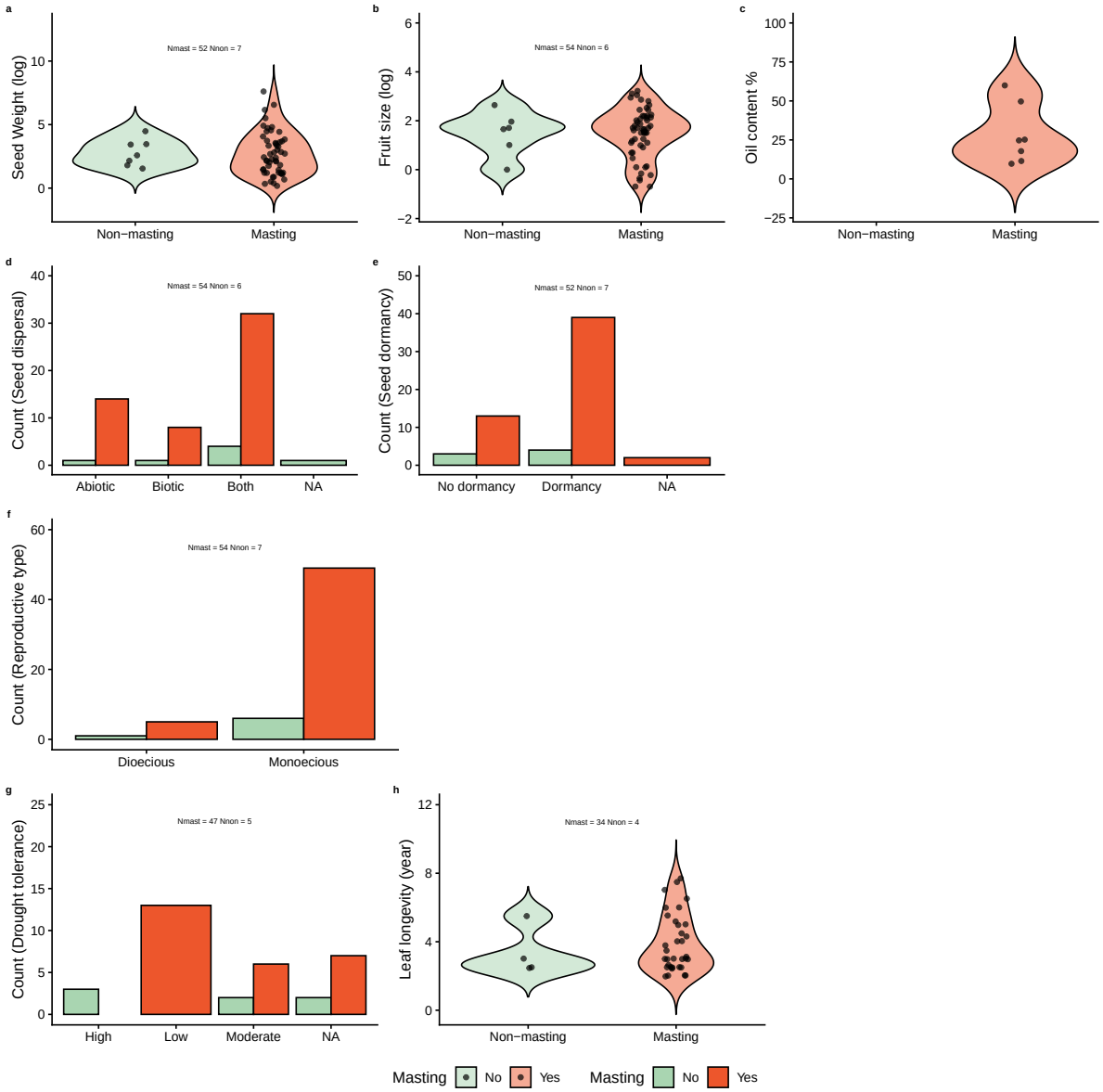

Figure 2: Relationships between plant traits (a) Seed weight; (b) Fruit size; (c) Oil content; (d) Dispersal mode; (e) Seed dormancy; (f) Pollination mode; (g) Reproductive mode; (h) Drought tolerance; (i) Leaf longevity and masting based on phylogenetic logistic regression for gymnosperm species. Raw data are shown for each trait: continuous traits are displayed as violin plots, and categorical traits as histograms. Overlaid model summaries are derived from phylogenetic logistic regression models fitted using phyloglm. For each model, we report sample size (N), regression coefficient (Estimate), phylogenetic signal parameter (α), and associated P-value. (a)-(e) are traits related to predation satiation hypothesis; (f), (g) are traits related to pollination coupling hypothesis; (h), (i) are traits related to resource matching hypothesis. Legends are shared across panels.

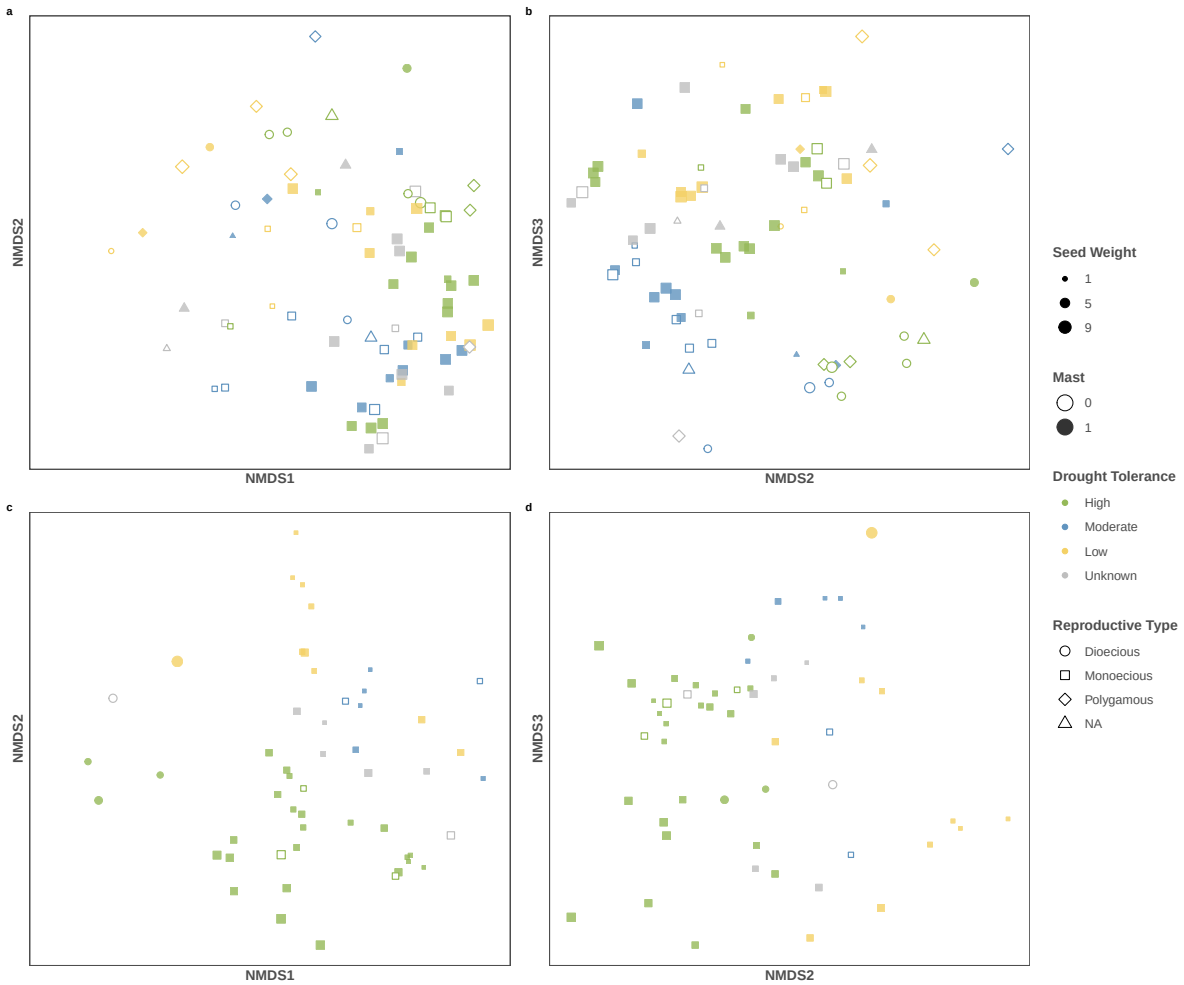

Figure 3: Non-metric multidimensional scaling (NMDS) of trait composition for angiosperm and conifer species. Each point represents a species, with position determined by NMDS axes. Points are colored by drought tolerance, filled by masting or not, shaped by reproductive type, sized by seed weight. Panels are arranged as (a) angiosperm NMDS1–NMDS2, (b) angiosperm NMDS2–NMDS3, (c) conifer NMDS1–NMDS2, and (d) conifer NMDS2–NMDS3.

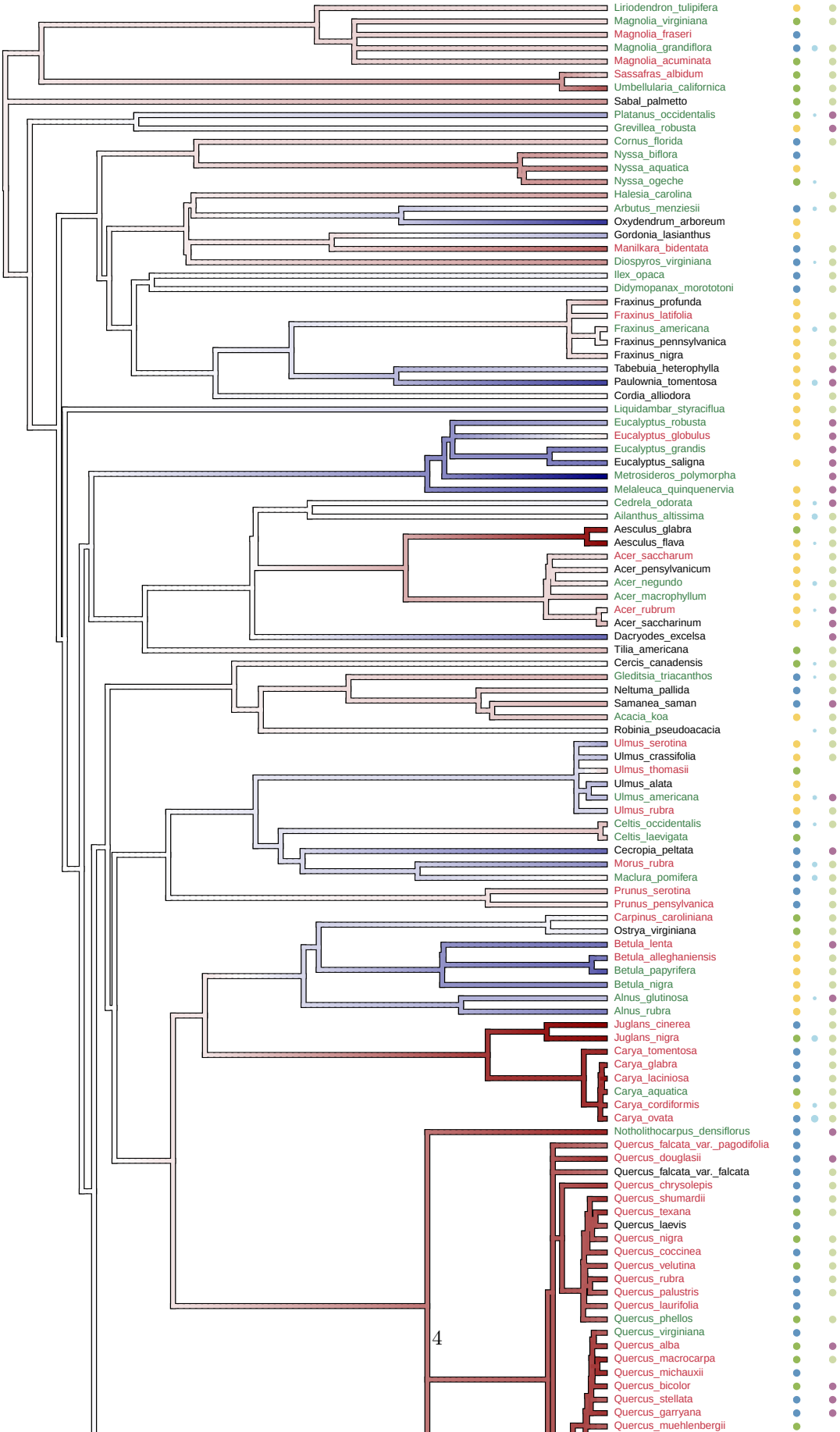
Liriodendron_tulipifera
Magnolia_virginiana
Magnolia_fraseri
Magnolia_grandiflora
Magnolia_acuminata
Sassafras_albidum
Umbellularia_californica
Sabal_palmetto
Platanus_occidentalis
Grevillea_robusta
Cornus_florida
Nyssa_biflora
Nyssa_aquatica
Nyssa_ogeche
Halesia_carolina
Arbutus_menziesii
Oxydendrum_arboreum
Gordonia_lasianthus
Manilkara_bidentata
Diospyros_virginiana
Ilex_opaca
Didymopanax_morototoni
Fraxinus_profunda
Fraxinus_latifolia
Fraxinus_americana
Fraxinus_pennsylvanica
Fraxinus_nigra
Tabebuia_heterophylla
Paulownia_tomentosa
Cordia_alliodora
Liquidambar_styraciflua
Eucalyptus_robusta
Eucalyptus_globulus
Eucalyptus_grandis
Eucalyptus_saligna
Metrosideros_polymorpha
Melaleuca_quinquenervia
Cedrela_odorata
Ailanthus_altissima
Aesculus_glabra
Aesculus_flava
Acer_saccharum
Acer_pensylvanicum
Acer_negundo
Acer_macrophyllum
Acer_rubrum
Acer_saccharinum
Dacryodes_excelsa
Tilia_americana
Cercis_canadensis
Gleditsia_triacanthos
Neltuma_pallida
Samanea_saman
Acacia_koa
Robinia_pseudoacacia
Ulmus_serotina
Ulmus_crassifolia
Ulmus_thomasii
Ulmus_alata
Ulmus_americana
Ulmus_rubra
Celtis_occidentalis
Celtis_laevigata
Cecropia_peltata
Morus_rubra
Maclura_pomifera
Prunus_serotina
Prunus_pensylvanica
Carpinus_caroliniana
Ostrya_virginiana
Betula_lenta
Betula_alleghaniensis
Betula_papyrifera
Betula_nigra
Alnus_glutinosa
Alnus_rubra
Juglans_cinerea
Juglans_nigra
Carya_tomentosa
Carya_glabra
Carya_laciniosa
Carya_aquatica
Carya_cordiformis
Carya_ovata
Notholithocarpus_densiflorus
Quercus_falcata_var._pagodifolia
Quercus_douglasii
Quercus_falcata_var._falcata
Quercus_chrysolepis
Quercus_shumardii
Quercus_texana
Quercus_laevis
Quercus_nigra
Quercus_coccinea
Quercus_velutina
Quercus_rubra
Quercus_palustris
Quercus_laurifolia
Quercus_phellos
Quercus_virginiana
Quercus_alba
Quercus_macrocarpa
Quercus_michauxii
Quercus_bicolor
Quercus_stellata
Quercus_garryana
Quercus_muehlenbergii
4

Figure 5: Phylogenetic tree for all Quercus species, Quercus species included in this study is color coded.

Group	Trait	Predictor	Estimate	SE	P	Phylo α	N
Gymnosperm	Seed dispersal	Biotic vs abiotic	-0.83	0.81	0.30	0.01	60
Gymnosperm	Seed dispersal	Both vs abiotic	-1.11	0.67	0.10	0.01	60
Gymnosperm	Seed dormancy	Dormant vs non-dormant	0.71	0.70	0.32	0.41	59
Gymnosperm	Reproductive type	Monoecious vs dioecious	-1.14	1.02	0.26	0.01	61
Gymnosperm	Seed weight (log)	logSeedWeight	-0.03	0.25	0.91	0.43	59
Gymnosperm	Fruit size (log)	logFruit	0.03	0.36	0.94	0.43	60
Gymnosperm	Seed size (log)	logSeedSize	-0.37	0.93	0.69	0.41	51
Gymnosperm	Leaf longevity (years)	Leaf longevity (years)	0.15	0.34	0.67	0.43	38
Gymnosperm	Drought tolerance	Low vs high drought tolerance	0.91	1.64	0.58	0.43	52
Gymnosperm	Drought tolerance	Moderate vs high drought tolerance	-1.56	1.06	0.14	0.43	52
Angiosperm	Dispersal mode	Biotic vs abiotic	0.55	0.53	0.30	0.06	93
Angiosperm	Dispersal mode	Both vs abiotic	0.55	0.57	0.33	0.06	93
Angiosperm	Pollination mode	Wind vs animal pollination	1.72	0.72	0.02	0.35	52
Angiosperm	Pollination mode	Wind vs animal pollination and animals	1.23	1.15	0.29	0.35	52
Angiosperm	Seed dormancy	Dormant vs non-dormant	0.03	0.50	0.95	0.10	85
Angiosperm	Reproductive type	Monoecious vs dioecious	0.90	0.68	0.18	0.11	89
Angiosperm	Reproductive type	Monoecious vs polygamous	-0.13	0.96	0.89	0.11	89
Angiosperm	Seed weight (log)	logSeedWeight	0.21	0.09	0.02	0.06	91
Angiosperm	Fruit size (log)	logFruit	0.08	0.30	0.80	0.17	88
Angiosperm	Seed size (log)	logSeedSize	0.24	0.30	0.43	0.08	69
Angiosperm	Oil content (%)	Oil content (%)	0.04	0.03	0.20	0.14	19
Angiosperm	Leaf longevity (years)	Leaf longevity (years)	0.26	0.95	0.78	0.01	16
Angiosperm	Drought tolerance	Low vs high drought tolerance	0.04	0.48	0.92	0.11	79
Angiosperm	Drought tolerance	Moderate vs high drought tolerance	-0.13	0.46	0.79	0.11	79

Table 1: Phylogenetic Generalized Linear Model Results

Group	Trait	Lambda
Angiosperm	Seed weight (Log)	0.95
Angiosperm	Fruit size (Log)	0.93
Angiosperm	Seed size (Log)	0.96
Angiosperm	Oil content	0.25
Angiosperm	Leaf longevity	1.02
Angiosperm	Dispersal mode	1.00
Angiosperm	Seed dormancy	1.00
Angiosperm	Pollination mode	1.00
Angiosperm	Reproductive type	1.00
Angiosperm	Drought tolerance	0.00
Angiosperm	Masting	0.00
Gymnosperm	Seed weight (Log)	0.97
Gymnosperm	Fruit size (Log)	0.90
Gymnosperm	Seed size (Log)	0.26
Gymnosperm	Oil content	0.00
Gymnosperm	Leaf longevity	0.90
Gymnosperm	Dispersal mode	1.00
Gymnosperm	Seed dormancy	1.00
Gymnosperm	Reproductive type	1.00
Gymnosperm	Drought tolerance	1.00
Gymnosperm	Masting	1.00

Table 2: Phylogenetic Signal (Pagel's $/lambda$) for All Traits